ASSIGNMENT 1: SCENARIO-BASED AIMS IMPLEMENTATION
Course: ISO/IEC 42001 Lead Implementer

The Scenario: ''AutoDrive Insurance AI''
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Date: 16 October 2026

STATEMENT OF SCOPE FOR THE ARTIFICIAL INTELLIGENCE MANAGEMENT SYSTEM (AIMS)
Document No: AD-AIMS-SCOPE-01 | Version: 1.0

1. PURPOSE:

The AIMS is established to govern the responsible development, deployment and operation of ClaimVision AI to ensure trustworthy, fair, accurate and compliant automated claims processing.

2. BOUNDARIES AND APPLICABILITY:

a) Systems Covered (In-Scope):
- ClaimVision AI Model (Computer Vision for damage assessment & cost estimation)
- ClaimVision Mobile App Module (photo capture feature)
- ClaimVision Backend API & Automated Payout Decision Engine (<$2000)
- AI Data Pipeline: acquisition, labeling, training, storage (500k photos)
- Model Monitoring & Performance Dashboard
- Human Flagging Logic for claims >$2000

b) Departments / Functions Covered:
- Data Science & AI Engineering (5 Data Scientists - Model Developers)
- IT & Cloud Operations (Deployment, Infrastructure)
- Claims Operations & Human Adjusters (Human-in-the-loop)
- Risk, Compliance & Legal (Regulatory & Fairness)
- Product Management (ClaimVision Owner)
- Data Governance & Procurement (Third-party data broker management)

c) AI Lifecycle Stages Covered (Clause 8):
- Data Acquisition & Data Quality Management
- AI System Design & Development
- Verification & Validation
- Deployment

- Operation & Continuous Monitoring
- Continual Improvement
- Responsible Decommissioning Planning

3. EXCLUSIONS WITH JUSTIFICATION:

- HR Recruitment AI & General Corporate IT: Excluded. Scope is limited to ClaimVision product only. Unrelated systems do not impact trustworthiness of claims AI.

- Physical Claims Inspection Process: Excluded. AIMS controls only the AI decision to approve/flag. Physical garage assessment is outside.

- Continuous Learning in Production: Excluded. ClaimVision v1.0 is a static model, does not perform online learning. Will be revisited if automated retraining is introduced.

TASK 2: AI RISK ASSESSMENT

RISK 1: DATA QUALITY & REPRESENTATIVENESS RISK

1. Risk Scenario: The 500,000 historical photos were purchased from a third-party data broker. No documentation of collection methods, consent, labeling accuracy, or distribution. Dataset may contain blurry, mislabeled, duplicate, or non-representative images (e.g., mostly newer cars). Leads to incorrect learning.

2. Potential Impact:

- To Organization: Inaccurate cost estimation, regulatory non-compliance, financial loss from overpayments, fail ISO 42001 audit.
- To Customer: Unfair claim denials or underpayment due to model failing on under-represented types.

3. Annex A Controls:

- A.7.1 Data Acquisition
- A.7.2 Data Quality
- A.7.3 Data Origin & Lineage

RISK 2: ALGORITHMIC BIAS - CAR COLOR / MODEL BIAS

1. Risk Scenario: Computer vision model may perform better on certain car colors (white/light) vs dark/black cars, or certain models (sedans vs SUVs) that were over-represented in broker data. Results in systematic bias.

2. Potential Impact:

- To Customer: Discriminatory outcome. Owners of black/dark cars consistently get lower estimates. Violates fairness and anti-discrimination laws.
- To Organization: Regulatory penalties, lawsuits, failure to meet fairness objectives, loss of trust.

3. Annex A Controls:

- A.8.2 Fairness & Non-discrimination
- A.6.2.4 Verification & Validation of AI System
- A.6.2.7 Documentation of Bias

RISK 3: AUTOMATION ERROR - LACK OF HUMAN OVERSIGHT FOR AUTO-PAYOUTS

1. Risk Scenario: For any claim <$2,000, ClaimVision automatically approves payout without human check. Adversary could submit manipulated photos (fraud), or model could have major error (e.g., estimates $800 for $5,000 damage). No human checks before money sent.

2. Potential Impact:

- To Organization: Direct financial loss due to fraud and overpayment. High customer appeals.
- To Customer: Underpayment, inability to repair vehicle, dissatisfaction.

3. Annex A Controls:

- A.6.2.5 Verification of AI System Output
- A.8.4 Monitoring & Evaluation of Performance in Operation
- A.8.5 Human Oversight & Human-in-the-loop

TASK 3: STATEMENT OF APPLICABILITY (SoA)

Control ID	Control Description	Included? (Y/N)	Justification
A.6.2	AI system development	Y	**Included.** AutoDrive is the AI Producer as stated in scenario. The AI model was developed entirely in-house by 5 data scientists. We have full control and responsibility for the design, development, training, and testing processes. Therefore, all controls under A.6.2 (design, development, V&V, documentation) must be implemented to ensure the model is fair robust and traceable for ISO 42001 certification.
A.6.6	AI system continuous learning	N	**Excluded.** Justification: ClaimVision v1.0 is described as a statically trained model using 500,000 historical photos. There is no evidence of online learning, reinforcement learning, or automatic retraining in production environment. The model does not learn continuously from new claim submissions. If AutoDrive plans to implement automated retraining in future, this control will be included. Manual periodic retraining is covered under A.6.2.
A.7.1	Data acquisition	Y	**Included.** Justification is critical. The model's training data (500k photos) was acquired from a third-party data broker, not collected directly. This introduces high risks related to data legality, consent, quality, IP rights, and bias. ISO/IEC 42001 requires controls to assess data sources, verify legitimacy, and document contracts/data sheets. This is mandatory for regulatory assurance.
A.8.1	Transparency of the AI system	Y	**Included.** The system makes automated financial decisions (payouts under $2,000) directly impacting customers without human intervention. Regulators and customers have the right to know that an AI is making the decision, its logic, limitations, and how to appeal. Transparency is a core principle of ISO 42001 and essential to achieve the Board's goal of assuring regulators that payouts are fair and accurate. Must implement explainability and customer notification.

TASK 4: CORRECTIVE ACTION PLAN - CLAUSE 10.1

INCIDENT: One month after launch, customer complains on social media that ClaimVision estimated $500 but mechanic bill was $2,500. Claims AI is "scam". Post goes viral. Investigation: AI never seen photos of this newly released EV model, failed to recognize expensive sensors behind bumper.

1. REACT: IMMEDIATE ACTION TO CONTAIN THE ISSUE [Clause 10.1 a & b]

a) Customer Containment:
- Immediately contact customer privately, apologize, escalate to Senior Human Adjuster.
- Pay difference $2,000 on goodwill basis after verifying genuine invoice.
- Request permission to use anonymized case for improvement.

b) System Containment:
- Temporary control: Suspend auto-approval for ALL newly released EV models (launched last 12 months). All claims for EV models 2024/2025/2026 must be flagged for human adjuster even if <$2000.
- Public holding statement: "We have heard concern, reviewing with priority, additional human checks for new EV models."
- Log as Nonconformity: NC-2026-01 in AIMS Nonconformity Register.

2. ROOT CAUSE ANALYSIS: WHAT PROCESS FAILED IN THE AIMS? [Clause 10.1 b]

5-Why Analysis:
Why 1: Why $500 vs $2500? Failed to recognize expensive sensors behind bumper.
Why 2: Why fail? Model never seen this specific new EV model.
Why 3: Why never seen? Training data was 500k historical photos from broker, only old models, no process to update for new models.
Why 4: Why no process? AIMS failed.

Failed Processes in AIMS:
- A.7.2 Data Quality & A.7.4 Data Relevance: No process to manage data drift and new car model coverage. No question of "What happens when 2026 car launches?"
- A.6.2.2 AI System Design - Intended Use: Defined use as "auto insurance damage" but did not define limitations - e.g., "Model only valid for models before 2023".
- A.8.4 Monitoring in Operation: No monitoring for performance on new car models. No trigger for "alert if car model with <100 samples".
- A.6.2.5 Verification & Validation: No testing for edge cases like new EVs with hidden ADAS sensors.

Final Root Cause Statement: The AIMS lacked lifecycle process for managing data representativeness for newly released vehicle models, and lacked operational boundary definition that would flag unknown models for human review. Nonconformity is systemic, not one-off.

3. CORRECTIVE ACTION: SYSTEMIC CHANGE TO PREVENT RECURRENCE [Clause 10.1 c,d,e,f]

A. Corrective Action Plan:

1. Update Risk Assessment [10.1 e]: Add new risk to AI Risk Register: "Risk of underestimation due to unseen new car models / technology (EVs, ADAS sensors)".

2. Implement New Controls:

- New Knowledge Base Control (A.7.1 & A.8.1): Create "Supported Vehicle Model List" database. Integration with automotive data provider (Edmunds API). If VIN / Model Year is <12 months old or not in training distribution, automatically flag for human adjuster regardless of amount.

- Data Update Process (A.7.5 & A.6.2.3): Establish Quarterly Data Refresh Procedure. Every quarter, acquire photos of new models launched. Minimum 200 photos per new model before auto-approval allowed. Until then, human-in-the-loop mandatory.

- Technical Fix - OOD Detector (A.6.2.4 & A.8.4): Implement Out-of-Distribution detector. If ClaimVision sees car shape it never saw, outputs low confidence (<85%). Any claim with confidence <85% goes to human even if $500. Retrain model to include sensor cost.

- Human Oversight Update (A.8.3): Revise SOP: All EV claims and all claims for vehicles launched within last 12 months require human validation.

- Transparency Update (A.8.1): Update App disclaimer: "Estimate by AI. For new models, estimate reviewed by human adjuster. You can request human review if you disagree."

B. Review Effectiveness [Clause 10.1 d]:
- After 30 days, audit: How many new EV models correctly flagged? Appeal rate? Error rate for new models dropped?
- KPIs: 0 auto-payouts for models with <200 training samples. 100% flagging of 2026 model year.
- Close NC-2026-01 only after evidence controls work.

C. Update AIMS Documentation [Clause 10.1 f]:
- Update Scope Statement to state limitation for new models.
- Update SoA, Risk Register, Improvement Log.
- Present to Management Review.

CONCLUSION: This incident proves Clause 10.1 works. We contained issue, paid customer fairly, stopped viral damage, found root cause was missing new-model management, and now have systemic fix so when 2027 cars launch, AI will ask for human help instead of guessing $500. This is what auditor wants to see - not perfect AI, but responsible management system that learns from failure.